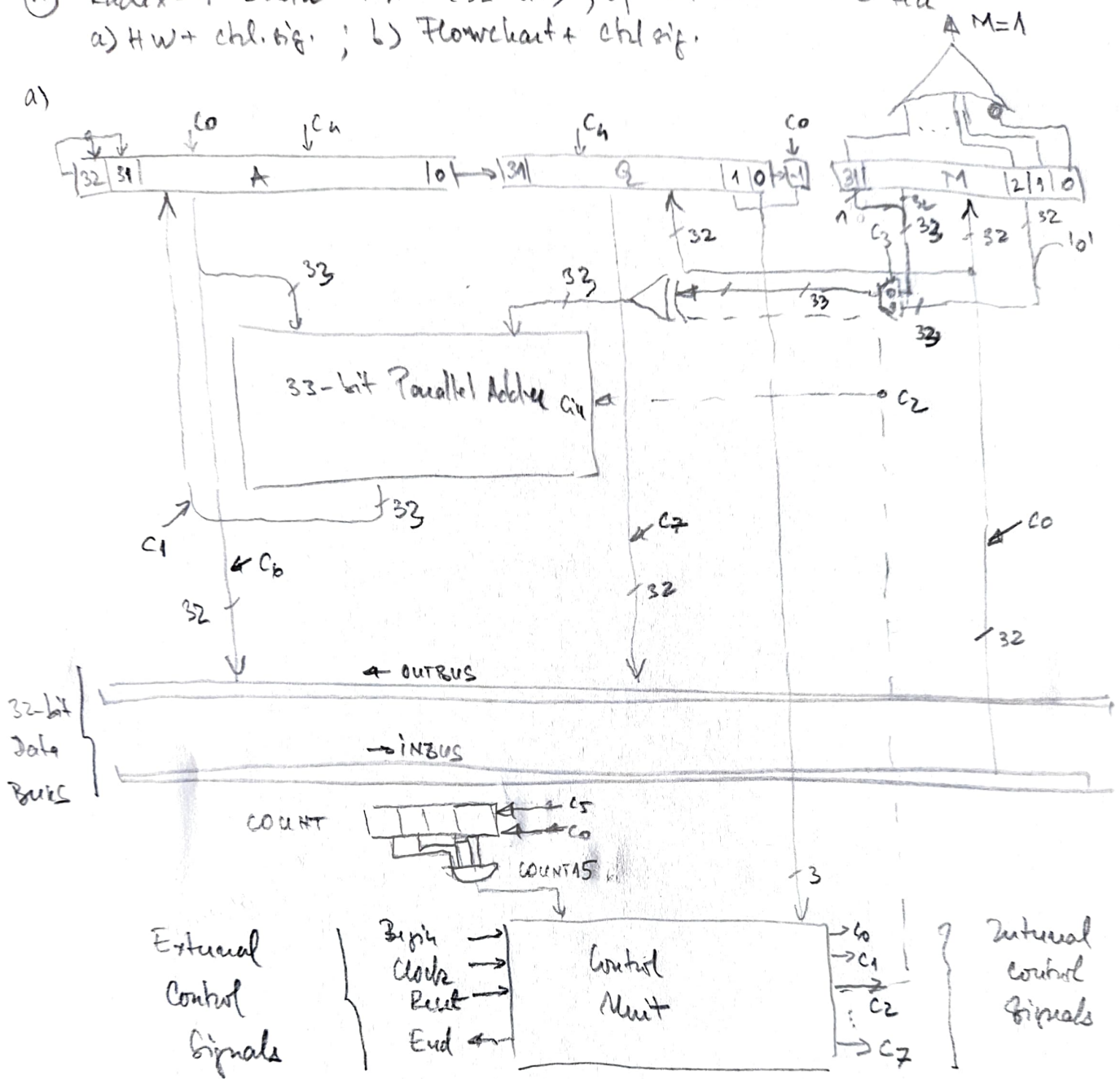


① Radix-4 Booth $\rightarrow X^2$ (32-bit); special case $X = (1)_{4u}$
 a) HW + chl. sig. ; b) Flowchart + chl. sig.

a)



Q b)

Start/stop ← Begin

↓ Begin

-C0

A := 0, COUNT := 0, Q[1] := 0,
M := INBUS, Q := INBUS;

M = 1

1

Q[1] = 1

001
010

C1

A := A + M;

011

C1, C2

A := A + 2M;

101

110

A := A - M;

100

C1, C2, C3

A := A - 2M;

000

111

-C4

A[30:0], Q := A, Q[31:2],
A[30] := A[31], A[31] := A[31];

COUNT = 1

0

COUNT := COUNT + 1;

C5

1

C6

OUTBUS := A;

C7

OUTBUS := Q;

Start/stop

② M

Type	T_{type}	CPI
ALU	50%	6
CMP	20%	5
B	20%	2
O	10%	4

$$f = 5 \text{ GHz}$$

$$IC = 10^6$$

$$\text{HW CPI}_{ALU} = 2$$

$$\text{CPI}_B = 1$$

$\Rightarrow$ Clock cycle time degradation = ?

$$\text{SW IC}_{ALU} = 5\%$$

$$\text{IC}_B = 2\%$$

$$\text{IC}_{CMP} = 1\%$$

HW > SW

a) CPU time
b) MIPS

Original M

$$\text{CPI}_M = 0.5 \times 6 + 0.2 \times 5 + 0.2 \times 2 + 0.1 \times 4 = 4.8$$

$$\text{CCT}_M = \frac{1}{5 \times 10^8 \text{ s}^{-1}} = 0.2 \times 10^{-9} \text{ s} = 0.2 \text{ ns}$$

$$\text{CPU time}_M = 10^6 \times 4.8 \times 0.2 \times 10^{-9} \text{ s} = 0.96 \times 10^{-3} \text{ s} = 0.96 \text{ ms}$$

$$\text{MIPS}_M = \frac{10^6}{0.96 \times 10^{-3} \times 10^6} = \frac{1000}{0.96} = 1041.66$$

SW

$$\text{IC}_{ALU_M} = 500\,000 = 5 \times 10^5$$

$$\text{IC}_{ALU_{SW}} = 4.75 \times 10^5$$

$$\text{IC}_{CMP_M} = 200\,000 = 2 \times 10^5$$

$$\text{IC}_{CMP_{SW}} = 1.98 \times 10^5$$

$$\text{IC}_{B_M} = 200\,000 = 2 \times 10^5$$

$$\text{IC}_{B_{SW}} = 1.96 \times 10^5$$

$$\text{IC}_{O_M} = 100\,000 = 1 \times 10^5$$

$$\text{IC}_{O_{SW}} = 1 \times 10^5$$

$$5 \times 10^5 \times 0.95 = 4.75 \times 10^5$$

$$2 \times 10^5 \times 0.99 = 1.98 \times 10^5$$

$$2 \times 10^5 \times 0.98 = 1.96 \times 10^5$$

$$\text{IC}_{SW} = 9.69 \times 10^5$$

$$\text{CPI}_{SW} = \frac{4.75 \times 6 \times 10^5 + 1.98 \times 5 \times 10^5 + 1.96 \times 2 \times 10^5 + 1 \times 4 \times 10^5}{9.69 \times 10^5}$$

$$= \frac{46.32}{9.69} = 4.78$$

$$\Rightarrow \text{CPU time}_{SW} = 9.69 \times 10^5 \times \frac{46.32 \times 10^5}{9.69 \times 10^5} \times 0.2 \text{ ns} = 9.264 \times 10^5 \times 10^{-9} \text{ s}$$

$$= 0.9264 \times 10^{-3} \text{ s} =$$

$$\text{MIPS}_{SW} = \frac{9.69 \times 10^5}{0.9264 \times 10^{-3} \times 10^6} = \frac{0.969}{0.9264 \times 10^{-3}} = 1045.984$$

$$= \frac{969}{0.9264} = 1045.984$$

HW $CPI_{ALU} = 4$ (HW + code la cei care au condusat $CPI_{ALU} = 5$
 și $CPI_B = 0$) și $CPI = 0$ nu repara.
 $CPI_{CMP} = 5$
 $CPI_B = 1$ $CCT_{HW} = 1.2 \times CCT_M$ (depende de $0, n$)
 $CPI_0 = 4$ IC_5 will not change

$$CPI_{HW} = 0.5 \times 4 + 0.2 \times 5 + 0.2 \times 1 + 0.1 \times 4 = 3.6$$

$$\begin{aligned}
 CPU_{time\ HW} &= 10^6 \times 3.6 \times (1+x) \times 0.2 \times 10^{-9} s \\
 &= 3.6 \times 0.2 \times (1+x) \text{ ms} = 0.72 \times (1+x) \text{ ms}
 \end{aligned}$$

$$CPU_{time\ HW} < CPU_{time\ SW}$$

$$0.72 \times (1+x) < 0.9264$$

$$0.72 + 0.72x < 0.9264 \Rightarrow 0.72x < 0.2064$$

$$\Rightarrow x < \frac{0.2064}{0.72} = 0.287$$

$$\boxed{x < 28.7\%}$$

$$\begin{aligned}
 MIPS_{HW} &= \frac{10^6}{0.72 \times (1+x) \times 10^{-3} \times 10^6} = \\
 &= \frac{10^3}{0.72(1+x)}
 \end{aligned}$$

$$MIPS_{HW} > MIPS_{SW} \Rightarrow \frac{10^3}{0.72(1+x)} > \frac{969}{0.9264}$$

$$\frac{1}{0.72(1+x)} > \frac{0.969}{0.9264} \Rightarrow \frac{1}{0.72(1+x)} > \frac{1}{0.956}$$

$$\Rightarrow 0.72(1+x) < 0.956$$

$$0.72 + 0.72x < 0.956 \Rightarrow 0.72x < 0.236$$

$$x < \frac{0.236}{0.72} = 0.328$$

$$\Rightarrow \boxed{x < 32.8\%}$$